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Studierichting: Informatica

Oefeningenreeks 4A nr. 48.5

$$\begin{aligned}\int \ln \left(\frac{x^2 + 1}{x^2 - 1} \right) dx &= x \ln \left(\frac{x^2 + 1}{x^2 - 1} \right) - \int \frac{x^2 - 1}{x^2 + 1} \cdot \frac{2x(x^2 - 1) - 2x(x^2 + 1)}{(x^2 - 1)^2} x dx \\&= x \ln \left(\frac{x^2 + 1}{x^2 - 1} \right) - \int \frac{-4x^2}{(x^2 + 1)(x - 1)(x + 1)} dx \\&\Rightarrow \frac{-4x^2}{(x^2 + 1)(x - 1)(x + 1)} = \frac{Ax + B}{x^2 + 1} + \frac{C}{x - 1} + \frac{D}{x + 1} \\&\Rightarrow A(i) + B = \frac{-4i}{i^2 - 1} = -2 \Rightarrow A = 0, B = -2 \\&\Rightarrow C = \frac{-4}{(1 + 1)(1 + 1)} = -1 \\&\Rightarrow D = \frac{-4}{(1 + 1)(-1 - 1)} = 1 \\&= x \ln \left(\frac{x^2 + 1}{x^2 - 1} \right) + \int \frac{1}{x - 1} dx - \int \frac{1}{x + 1} dx + 2 \int \frac{1}{x^2 + 1} dx \\&= x \ln \left(\frac{x^2 + 1}{x^2 - 1} \right) + \ln |x - 1| - \ln |x + 1| + 2 \operatorname{Bgtan} x + c\end{aligned}$$

References